

ISABELLE STRATTON

LinkedIn

EDUCATION

Carnegie Mellon University	M.S. Sustainable and Regenerative Design	May 2027
University of Florida	B.S. Mathematics	August 2023
University of Florida	B.F.A. Fine Arts	August 2023

TECHNICAL SKILLS

Building analysis	Revit (modeling and energy analysis), IESVE (CFD and energy modelling), EnergyPlus, eQUEST, ClimateStudio, Climate Consultant
Computational design	Grasshopper, Python (parametric workflows and scripting)
3D modeling	Rhino, SolidWorks
Geospatial	ArcGIS ecosystem
Visualization	Adobe Creative Suite, Microsoft Office

EXPERIENCE

Carnegie Mellon University	2026–Present
Research Assistant	Pittsburgh, PA

Design and prototype sustainable materials from biomaterials and construction waste within a circular-economy framework. Characterize and test bio-based and waste-derived materials to evaluate their performance and environmental impact.

The University of Florida	2025–2026
GIS Analyst	Gainesville, FL

Worked with Dr. Kotryna Klizentyte to build an ArcGIS tool that maps parcel-level ecosystem service values for conservation decision making. Processed spatial data on parcels, canopy, wetlands and soils, then joined economic valuation models to the GIS layers to produce per-parcel outputs. Built a unified geodatabase and an interactive web map for county stakeholders.

Behavioral Health Services North	2024–Present
Data Analyst	Plattsburgh, NY

Deliver diagnostic and predictive analyses of healthcare needs that meet CCBHC reporting standards. Built a risk-prediction model with multivariate statistics and a Random Forest classifier. Manage databases in Excel and SQL and analyze protected health information in full HIPAA compliance.

Hulcr Lab	2023
Laboratory Technician	Gainesville, FL

Worked with Christopher Marais on PCA and K-means clustering for an AI-based bark beetle classifier. Trained new technicians in sample processing and photography, and operated a macro-camera to collect AI data.

PUBLICATIONS

Decision-making tool to value ecosystem services on potential land acquisitions in Hillsborough County, Florida	In preparation
Isabelle C. Stratton, Kotryna Klizentyte	Gainesville, FL

Writing a publication on the Hillsborough County ecosystem service valuation project, which I will present at the Tampa Bay Innovation Summit.

Progress in Developing a Bark Beetle Identification Tool	2025
G. Christopher Marais, Isabelle C. Stratton, Andrew J. Johnson, Jiri Hulcr	Gainesville, FL

The paper describes a tool that identifies bark beetles, invasive species that damage forests globally and resist identification because their morphology is uniform.

PROJECTS

Generational Relief in Prosthetics

2023

One of six members of the 3D-Printed Assistive Devices Division. Converted digital renderings of assistive prosthetics into 3D models for recipients, repaired 3D printers and helped adapt toys for movement-restricted children.

Touch-Activated Warming Sandbox

2022

Wrote Arduino code for touch sensors and 12-volt heating elements and fixed voltage-regulation issues. Built a working prototype for educational and therapeutic use.

AWARDS

Robert Long Mathematics Essay Contest, Honorable Mention for an essay on a-periodic tilings in Islamic mosques (2023)

Neilson Scholarship, with a preparator internship at the Harn Museum of Art; one of the youngest recipients (2020)

American Visions Award, Scholastic Art and Writing, for illustration; one awarded per region nationally (2019)

National Gold Medalist, Scholastic Art and Writing, illustration and mixed media (2018, 2019)

Bright Futures Scholar, full undergraduate tuition (2019–2023)

Raimondi Family Photography Scholarship (2019)

Jason Budjinski Memorial Scholarship, academic excellence (2019)